

实验 8: PyTorch 入门与图像分类

Computer Vision 上机练习任务说明

一、实验目标

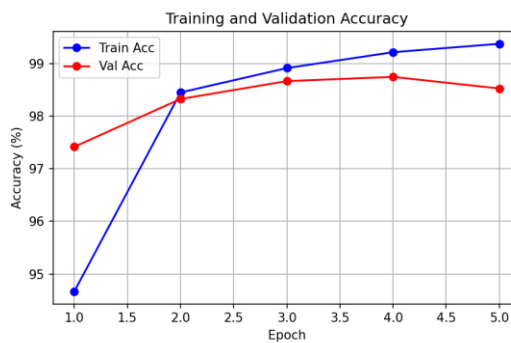
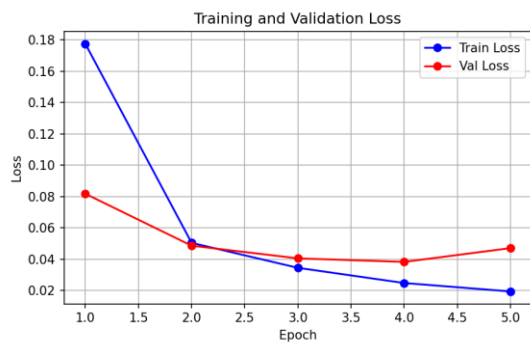
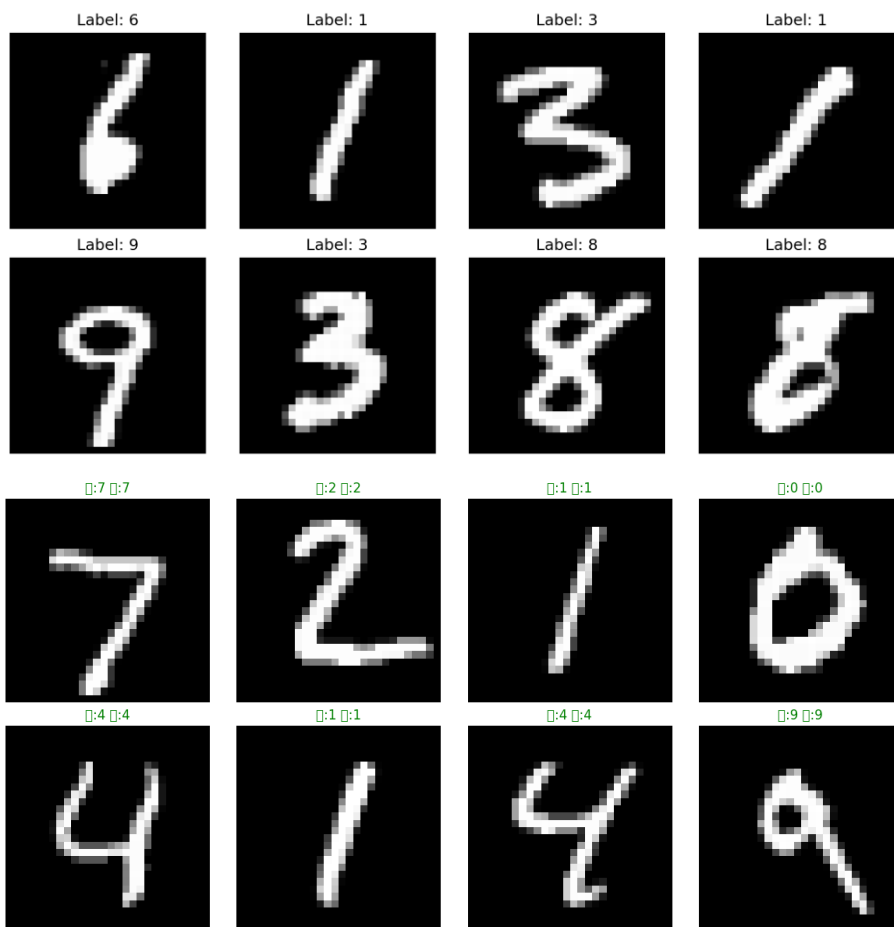
本实验要求使用 *PyTorch* 完成一个简单的图像分类任务，完整体验深度学习图像分类的基本流程：

数据加载 → 模型定义 → 训练 → 验证 → 测试 → 结果分析

完成实验后，学生应能够理解深度学习图像分类任务的基本组成，并能够独立运行一个简单的 *CNN* 图像分类实验。

二、实验任务

```
Epoch 1: Train Loss=0.1740, Acc=94.81% | Val Loss=0.0690, Acc=97.84%
Epoch 2: Train Loss=0.0483, Acc=98.54% | Val Loss=0.0562, Acc=98.21%
Epoch 3: Train Loss=0.0336, Acc=98.94% | Val Loss=0.0477, Acc=98.61%
Epoch 4: Train Loss=0.0245, Acc=99.22% | Val Loss=0.0372, Acc=99.02%
Epoch 5: Train Loss=0.0174, Acc=99.41% | Val Loss=0.0461, Acc=98.74%
```



任务 8：结果分析

- 在实验报告中回答：训练 *loss* 是否随着 *epoch* 增加而下降？
- 验证 *accuracy* 是否随着训练逐渐提升？
- 训练 *accuracy* 和验证 *accuracy* 是否存在明显差距？
- 如果存在明显差距，可能是什么原因？
- 哪些数字或类别更容易被错误分类？
- *MNIST* 和 *CIFAR-10* 哪个更难？为什么？

三、进阶任务

完成基础任务后，可以选择完成以下进阶任务。

进阶任务 1：修改网络结构

```
=====
训练模型：基线模型（2卷积层）
参数量：421,642
=====
Epoch 1: Train Loss=0.1817, Acc=94.58% | Val Loss=0.0633, Acc=97.95%
Epoch 2: Train Loss=0.0495, Acc=98.45% | Val Loss=0.0526, Acc=98.25%
Epoch 3: Train Loss=0.0334, Acc=98.90% | Val Loss=0.0412, Acc=98.73%
Epoch 4: Train Loss=0.0253, Acc=99.20% | Val Loss=0.0482, Acc=98.43%
Epoch 5: Train Loss=0.0185, Acc=99.42% | Val Loss=0.0370, Acc=98.90%

测试集准确率：98.96%

=====
训练模型：加深模型（3卷积层）
参数量：390,410
=====
Epoch 1: Train Loss=0.1826, Acc=94.15% | Val Loss=0.0632, Acc=97.85%
Epoch 2: Train Loss=0.0445, Acc=98.61% | Val Loss=0.0614, Acc=98.18%
Epoch 3: Train Loss=0.0312, Acc=99.03% | Val Loss=0.0525, Acc=98.45%
Epoch 4: Train Loss=0.0243, Acc=99.26% | Val Loss=0.0481, Acc=98.62%
Epoch 5: Train Loss=0.0199, Acc=99.32% | Val Loss=0.0334, Acc=99.05%

测试集准确率：99.10%

=====
训练模型：高级模型（BN+Dropout）
参数量：422,986
=====
Epoch 1: Train Loss=0.1425, Acc=96.25% | Val Loss=0.0469, Acc=98.65%
Epoch 2: Train Loss=0.0546, Acc=98.42% | Val Loss=0.0359, Acc=98.90%
Epoch 3: Train Loss=0.0420, Acc=98.77% | Val Loss=0.0363, Acc=98.89%
Epoch 4: Train Loss=0.0359, Acc=98.92% | Val Loss=0.0281, Acc=99.18%
Epoch 5: Train Loss=0.0300, Acc=99.11% | Val Loss=0.0366, Acc=99.03%
```

模型性能对比总结			
模型名称	参数量	测试准确率	
基线模型（2卷积层）	421,642	98.96	%
加深模型（3卷积层）	390,410	99.10	%
高级模型（BN+Dropout）	422,986	99.03	%

- 1. 加深模型（3 层卷积）相比基线模型，准确率可能提升
- 2. 加入 *BatchNorm* 和 *Dropout* 可以缓解过拟合
- 3. 参数量增加会带来更好的表达能力，但也需要更多训练时间

进阶任务 2：比较不同优化器

- 比较至少两种优化器，例如 *SGD* 和 *Adam*。

优化器性能对比总结		
优化器 (学习率)	测试准确率	
SGD (lr=0.01)	98.91	%
SGD (lr=0.001)	97.45	%
Adam (lr=0.01)	97.21	%
Adam (lr=0.001)	98.73	%
RMSprop (lr=0.001)	99.05	%

- 记录不同优化器、*learning rate* 和 *test accuracy*。

```
=====
优化器: SGD, 学习率: 0.01
=====
Epoch 1: Train Loss=0.2567, Acc=92.17% | Val Loss=0.0742, Acc=97.76%
Epoch 2: Train Loss=0.0573, Acc=98.22% | Val Loss=0.0492, Acc=98.46%
Epoch 3: Train Loss=0.0400, Acc=98.77% | Val Loss=0.0472, Acc=98.57%
Epoch 4: Train Loss=0.0300, Acc=99.11% | Val Loss=0.0418, Acc=98.80%
Epoch 5: Train Loss=0.0241, Acc=99.26% | Val Loss=0.0404, Acc=98.82%

测试集准确率: 98.91%

=====
优化器: SGD, 学习率: 0.001
=====
Epoch 1: Train Loss=0.9641, Acc=76.70% | Val Loss=0.2946, Acc=91.26%
Epoch 2: Train Loss=0.2254, Acc=93.35% | Val Loss=0.1867, Acc=94.40%
Epoch 3: Train Loss=0.1488, Acc=95.47% | Val Loss=0.1279, Acc=96.18%
Epoch 4: Train Loss=0.1111, Acc=96.78% | Val Loss=0.1026, Acc=96.97%
Epoch 5: Train Loss=0.0911, Acc=97.33% | Val Loss=0.0957, Acc=97.06%

测试集准确率: 97.45%

=====
优化器: Adam, 学习率: 0.01
=====
Epoch 1: Train Loss=0.1952, Acc=93.98% | Val Loss=0.1108, Acc=96.58%
Epoch 2: Train Loss=0.0932, Acc=97.18% | Val Loss=0.0952, Acc=97.09%
Epoch 3: Train Loss=0.0795, Acc=97.66% | Val Loss=0.0797, Acc=97.37%
Epoch 4: Train Loss=0.0704, Acc=97.86% | Val Loss=0.0771, Acc=97.72%
Epoch 5: Train Loss=0.0717, Acc=97.80% | Val Loss=0.0890, Acc=97.29%

测试集准确率: 97.21%
```

```

=====
优化器: Adam, 学习率: 0.001
=====
Epoch 1: Train Loss=0.1785, Acc=94.61% | Val Loss=0.0568, Acc=98.35%
Epoch 2: Train Loss=0.0510, Acc=98.44% | Val Loss=0.0514, Acc=98.50%
Epoch 3: Train Loss=0.0335, Acc=98.93% | Val Loss=0.0437, Acc=98.72%
Epoch 4: Train Loss=0.0251, Acc=99.20% | Val Loss=0.0440, Acc=98.71%
Epoch 5: Train Loss=0.0181, Acc=99.40% | Val Loss=0.0462, Acc=98.62%

测试集准确率: 98.73%

=====
优化器: RMSprop, 学习率: 0.001
=====
Epoch 1: Train Loss=0.1732, Acc=94.80% | Val Loss=0.0584, Acc=98.33%
Epoch 2: Train Loss=0.0494, Acc=98.42% | Val Loss=0.0482, Acc=98.63%
Epoch 3: Train Loss=0.0330, Acc=98.96% | Val Loss=0.0512, Acc=98.41%
Epoch 4: Train Loss=0.0239, Acc=99.25% | Val Loss=0.0516, Acc=98.36%
Epoch 5: Train Loss=0.0179, Acc=99.44% | Val Loss=0.0379, Acc=98.93%

测试集准确率: 99.05%

```

- 简要分析不同优化器对训练效果的影响。
 1. Adam 通常比 SGD 收敛更快, 准确率更高
 2. 学习率影响: $lr=0.001$ 通常比 $lr=0.01$ 更稳定
 3. SGD 需要 momentum 辅助才能达到较好效果
 4. 对于 MNIST 这类简单任务, 各优化器差异不大

进阶任务 3: 比较 MNIST 和 CIFAR-10

1. 测试准确率对比:

- MNIST: 99.03%

- CIFAR-10: 68.24%

2. CIFAR-10 比 MNIST 更难的原因:

- 彩色图像 (3 通道) vs 灰度图像 (1 通道)
- 图像内容更复杂 (物体 vs 数字)
- 同类变化更大 (猫有很多种样子)
- 背景干扰更多
- 需要更强的模型和更多训练时间

3. 训练时间:

- *CIFAR-10* 每个 *epoch* 更慢（图像更大、更复杂）